HOMEWORK 9.2 · UNIT 9 · NC.8.G.9

Volume Day 2

Targets: G.9-7, G.9-3, G.9-5, G.9-4, G.9-2, G.9-8, G.9-9, G.9-10

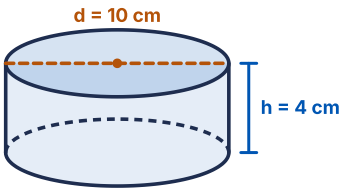
Name _____ Date _____ Block _____

Remember: Cylinder $V = \pi r^2 h$ · Cone $V = \frac{1}{3}\pi r^2 h$ · Sphere $V = \frac{4}{3}\pi r^3$. Given the diameter? Find $r = d \div 2$ first. Missing h or r? Fill in what you know, then solve. **Ex: cylinder** $V = 72\pi$, $r = 3$: $\pi(3)^2 h = 72\pi \rightarrow 9h = 72 \rightarrow h = 8$.

SCORE 3.0 Approaching · the basics

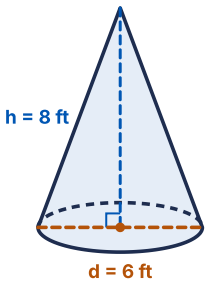
Targets: G.9-7, G.9-3, G.9-5, G.9-4, G.9-2

1 Find the volume. Answer in terms of π. G.9-7

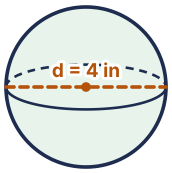


(A) 100π cm^3
(B) 400π cm^3
(C) 40π cm^3
(D) 20π cm^3

2 Find the volume of the cone. Use 3.14 for π. Round to the nearest tenth. G.9-3,G.9-5



3 Find the volume of the sphere. Answer in terms of π. G.9-4



4 Which formula finds the volume of a basketball? REVIEW 9.1 G.9-4

(A) $V = \pi r^2 h$
(B) $V = \frac{1}{3}\pi r^2 h$
(C) $V = \frac{4}{3}\pi r^3$

5 A cylinder has r = 3 m and h = 7 m. Find its volume. Answer in terms of π. REVIEW 9.1 G.9-2

SCORE 4.0 Proficient · on target Targets: G.9-8, G.9-7

6 A cylinder holds 98π m^3. Its radius is 7 m. Find its height. G.9-8

7 **Ty** made a mistake. Circle it. Explain it. Fix it. G.9-7

TY'S WORK

Cone: $d = 6$ cm, $h = 10$ cm

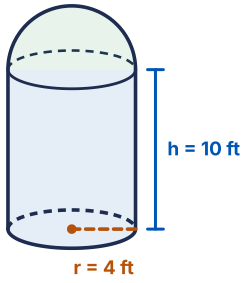
$V = \frac{1}{3}\pi(6)^2(10)$

$V = 120\pi$ cm^3

8 A gumball globe is a sphere with a diameter of 10 in. How much does it hold? Use 3.14 for π. Round to the nearest tenth. G.9-7

SCORE 5.0 Mastery · go deeper Targets: G.9-9, G.9-10

9 A grain silo is a cylinder with a hemisphere on top. Find its volume. Answer in terms of π. Then round. Show each step. G.9-9



Cylinder = _____ . Hemisphere = _____ . Total = _____ .

10 Cone A has r = 3 and h = 4. Cone B has r = 6 and h = 4. How many times more does B hold? Explain why. G.9-10

B holds _____ times as much because _____ .